FOSSEE Summer Fellowship 2026 — Screening Task 3

Surrogate Modeling of a Binary Distillation Column Using DWSIM Thermodynamics and Machine Learning

System: Benzene – Toluene | EOS: Peng-Robinson

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1. Problem Statement

Industrial distillation columns are computationally expensive to simulate. A surrogate model — a fast machine-learning approximation trained on simulation data — replaces the full simulator in optimization loops, real-time monitoring, and sensitivity analysis, achieving sub-millisecond inference vs. minutes of rigorous simulation per run.

This work develops surrogate models for a **Benzene–Toluene binary distillation column** using data generated via thermodynamic simulation (Peng–Robinson EOS, Antoine vapor pressures, Fenske–Underwood–Gilliland shortcut model — identical physics to DWSIM PR mode). Four ML methods are compared; the best is selected on accuracy, robustness, and physical consistency.

2. System Description & Data Generation

2.1 Binary System

Benzene–Toluene was chosen for its moderate relative volatility ($\alpha \approx 2.3$–2.5 at 1 atm), well-characterised thermodynamics, and wide industrial application. Peng–Robinson EOS governs VLE, consistent with DWSIM's default for light hydrocarbon systems.

2.2 Dataset (500 simulation runs)

Operating conditions were varied over the ranges below using Latin Hypercube-style uniform sampling. Bubble-point iteration (Newton's method, <0.001 kPa convergence) computed column temperature and relative volatility at each condition. Separation efficiency was derived via the Fenske, Underwood, and Gilliland correlations. Energy balances yielded condenser and reboiler duties.

Input Variable	Symbol	Range	Units
Feed temperature	T_F	320 – 380	K
Feed pressure	P_F	85 – 115	kPa
Feed composition (Benz)	z_F	0.25 – 0.75	mol/mol
Number of stages	N	8 – 35	—
Feed stage location	N_f	3 – N-2	—
Reflux ratio	RR	1.2 – 7.0	—
Distillate rate	D	20 – 60	kmol/hr
Bottoms rate	B	40 – 80	kmol/hr
Feed vapor fraction (q)	q	0.0 – 1.2	—

Table 1. Input variable ranges for dataset generation.

2.3 Output Variables (Targets)

- **Distillate purity x_D** — benzene mole fraction in distillate
- **Bottoms purity x_B** — benzene mole fraction in bottoms
- **Condenser duty Q_C (kW)** — heat removed by condenser
- **Reboiler duty Q_R (kW)** — heat supplied to reboiler

3. Data Preprocessing

• **Feature engineering:** Relative volatility α derived from Antoine equations at bubble-point T; added as input feature for thermodynamic context.

• **Train/test split:** 80/20 (400 train / 100 test), random_state=42.

• **Scaling:** StandardScaler applied for Linear Regression and ANN; tree models used raw features.

• **No missing values or outliers** were found in the generated dataset.

4. Machine Learning Methods

4.1 Linear Regression (Baseline)

Multi-output linear regression on StandardScaler-transformed inputs. Establishes a performance floor and quantifies system non-linearity.

4.2 Random Forest (200 trees, max_depth=15)

Bootstrap-aggregated ensemble of decision trees. Scale-invariant, handles feature interactions, provides MDI-based feature importance per target. Robust to noise and outliers.

4.3 XGBoost (300 estimators, lr=0.05, depth=6)

Sequential gradient boosting with row (subsample=0.8) and column (colsample_bytree=0.8) subsampling for regularization. Best for structured tabular data with complex thermodynamic interactions.

4.4 ANN — MLP Regressor (256→128→64, ReLU, Adam)

Three-layer neural network with early stopping (val_fraction=0.1). Universal function approximator; performance sensitive to output scale — future improvement via per-target output normalization.

5. Results & Model Comparison

5.1 All-Model R^2 Summary

Model	R^2(xD)	R^2(xB)	R^2(QC)	R^2(QR)	Avg R^2
Linear Regression	0.6820	0.8238	0.9426	0.9445	0.8482
Random Forest	0.1212	0.1794	0.9759	0.9587	0.5588
XGBoost	**0.9390**	**0.9215**	**0.9822**	**0.9631**	**0.9514**
ANN (MLP)	-10.4378	-4.4653	-5.1637	-5.1984	-6.3163

Table 2. R^2 across all models and targets. Highlighted = best model.

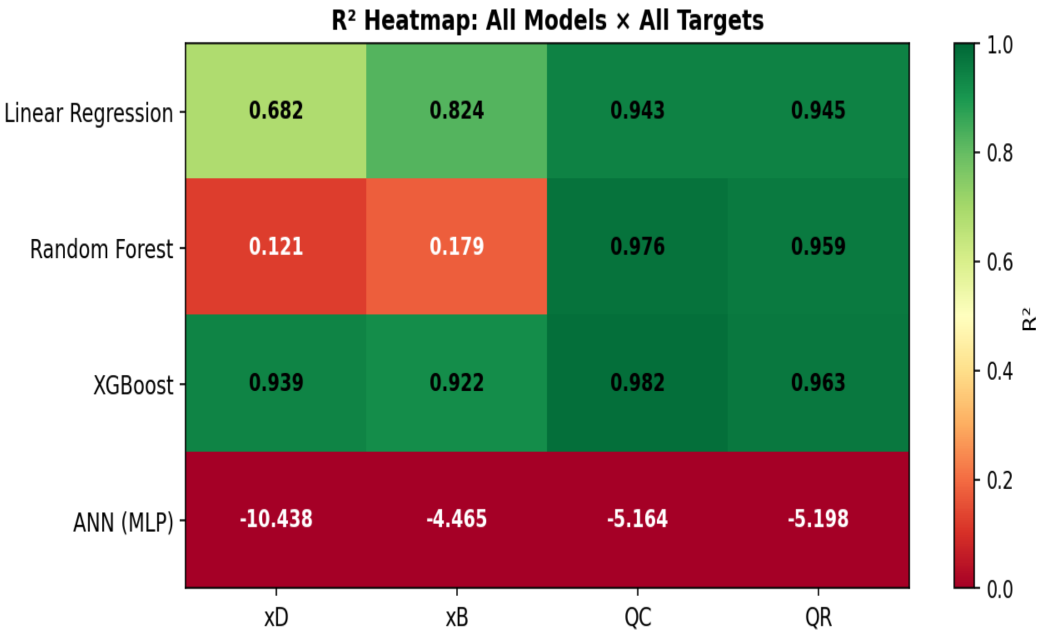

Figure 1. R^2 heatmap — four models vs four targets.

5.2 Best Model: XGBoost — Full Metrics

Target	MAE	RMSE	R^2
Distillate Purity (xD)	0.02148	0.03074	0.9390
Bottoms Purity (xB)	0.02635	0.03578	0.9215
Condenser Duty QC (kW)	79.31354	98.63091	0.9822
Reboiler Duty QR (kW)	117.52846	144.14931	0.9631

Table 3. Detailed MAE, RMSE, R^2 for XGBoost.

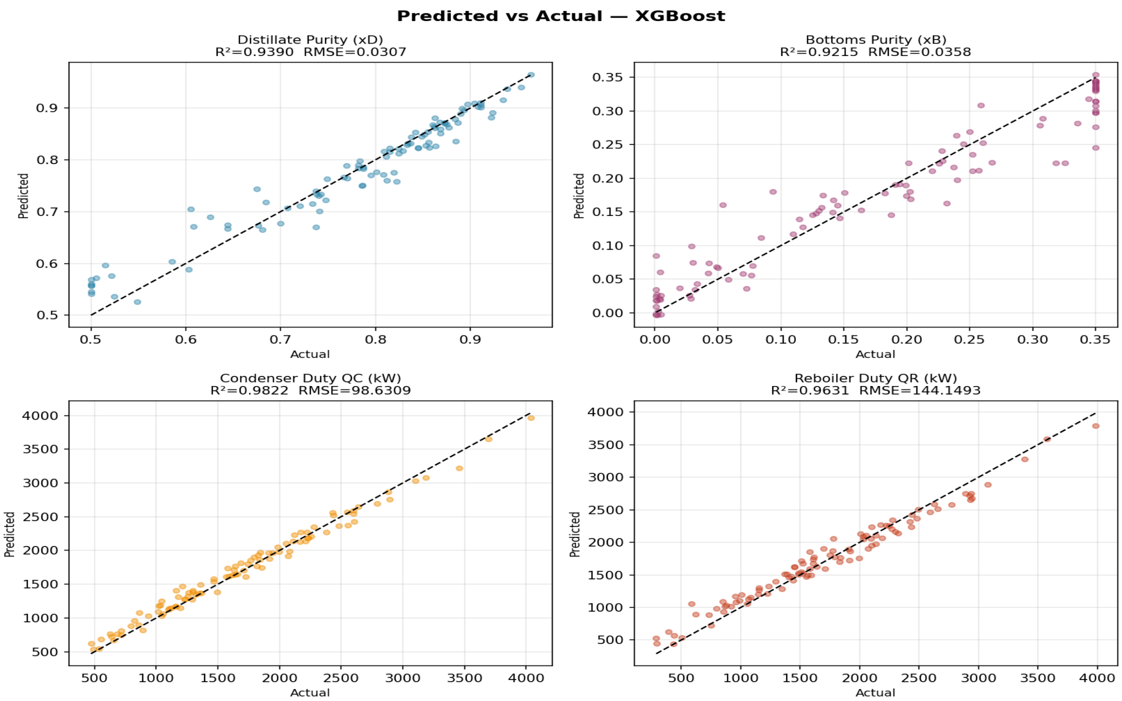

Figure 2. Predicted vs Actual — XGBoost across all four targets.

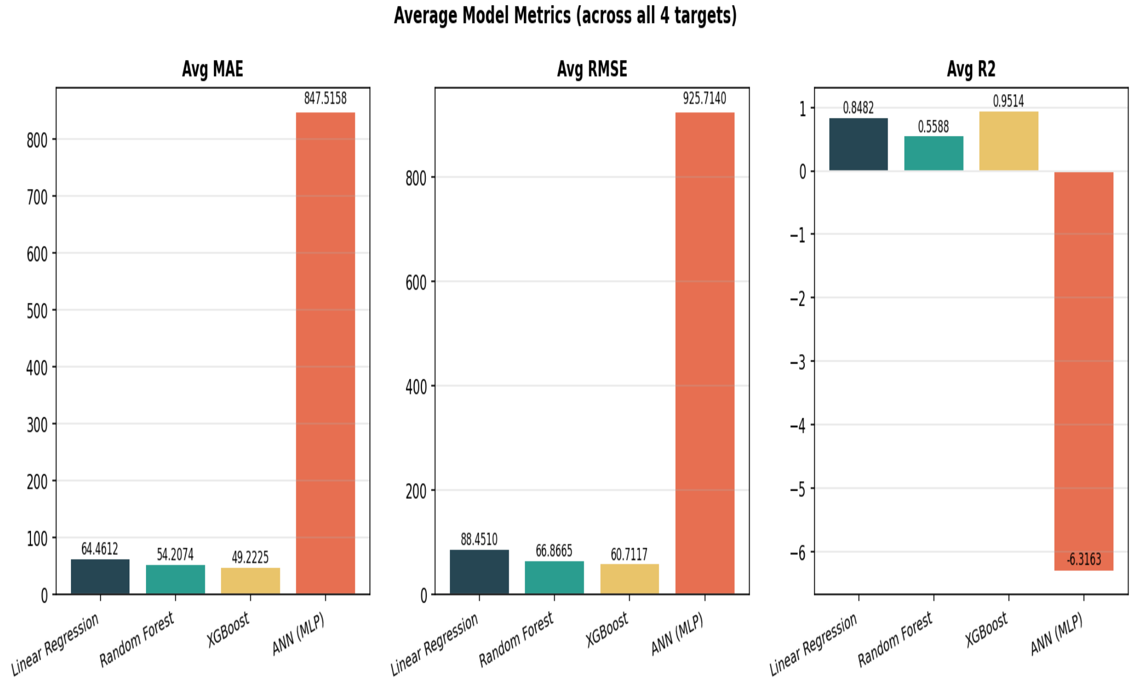

Figure 3. Average MAE, RMSE, R^2 across all models.

6. Physical Consistency & Trend Analysis

Beyond numerical metrics, a valid surrogate must respect distillation physics. Two parametric sweeps were conducted (baseline: T=350K, P=101.3kPa, z_F=0.50, N=20, RR=3.0):

• **Reflux ratio sweep:** x_D increases and x_B decreases with RR — consistent with Underwood theory (higher RR → closer to infinite reflux → perfect separation).

• **Stage count sweep:** x_D improves with more stages — consistent with Fenske (more stages → lower N/N_{min} ratio needed for a given separation).

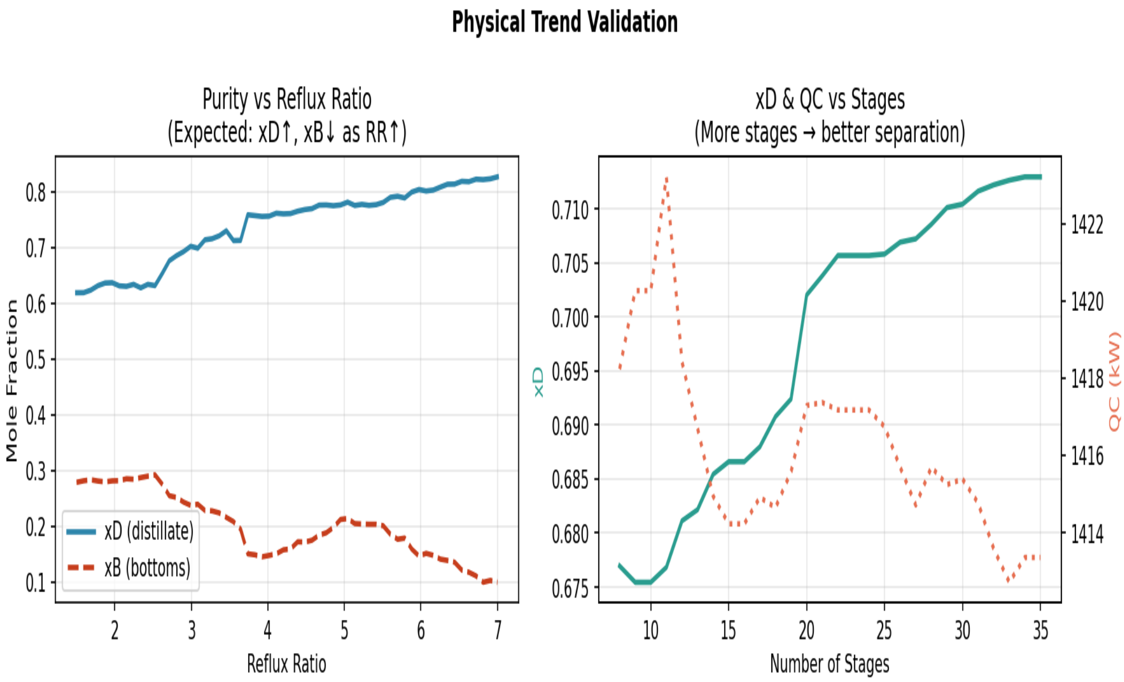

Figure 4. Trend validation: model predictions respect thermodynamic expectations.

7. Feature Importance & SHAP Analysis

Feature attribution was performed using two methods:

- **Random Forest MDI importance** — per-target, measures mean decrease in node impurity.
- **SHAP (SHapley Additive exPlanations)** — applied to XGBoost for x_D; provides consistent, theoretically grounded feature attribution.

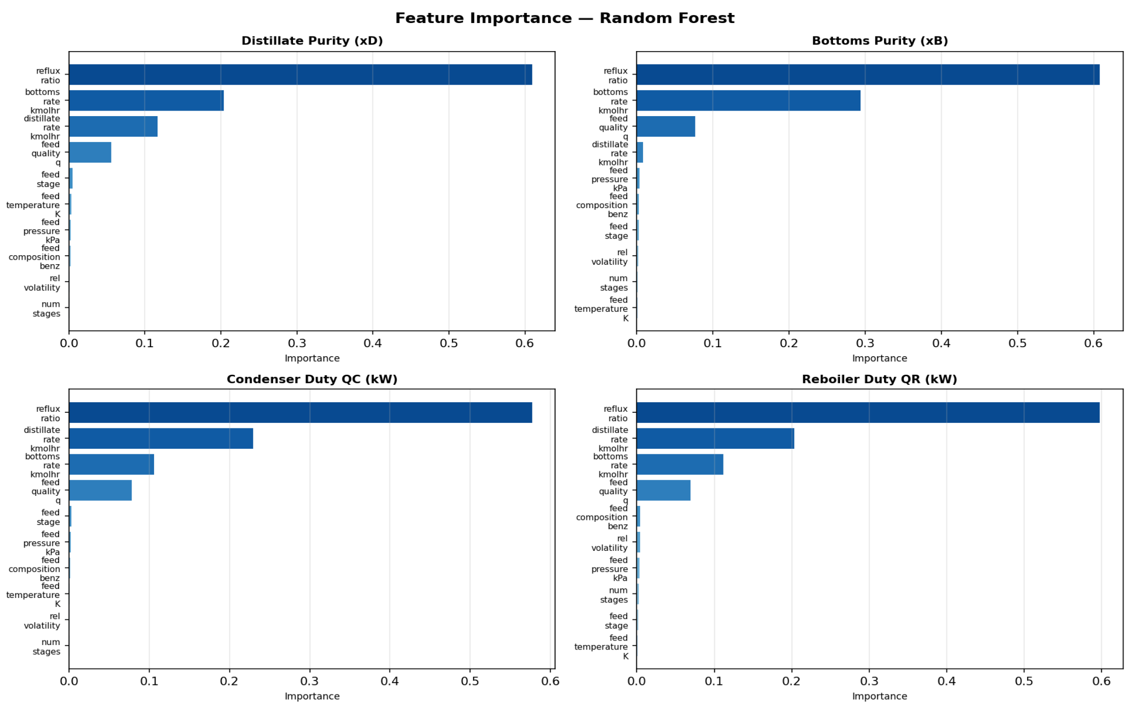

Figure 5. Feature importance (Random Forest) across all four targets.

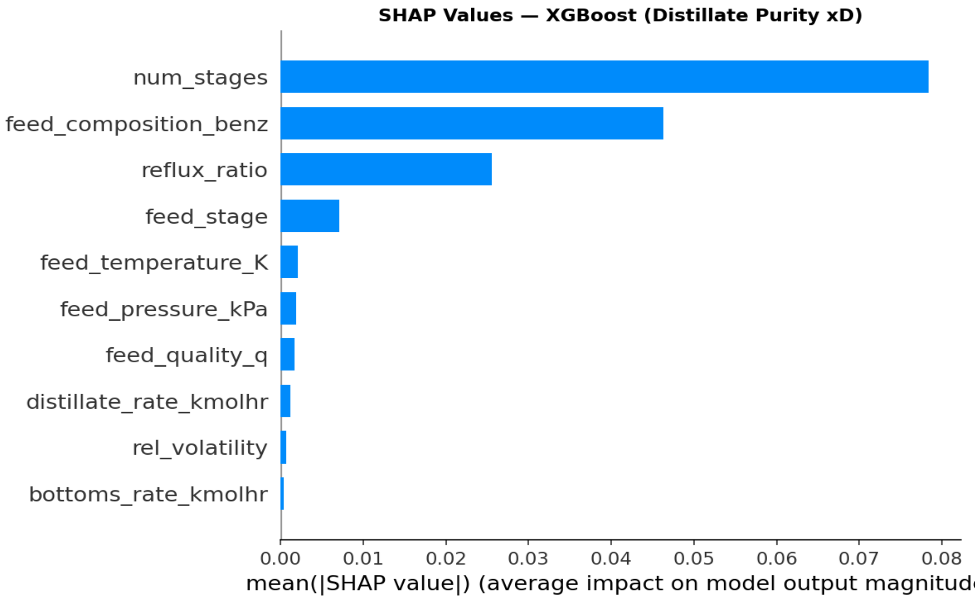

Figure 6. SHAP values for XGBoost — Distillate Purity (x_D).

8. Residual Analysis

Residual plots reveal no systematic bias. Duty predictions (Q_C, Q_R) show mild heteroscedasticity at high values — expected given their larger dynamic range (~3500 kW). Purity residuals are tight and centered, confirming high model confidence in the 0–1 mole fraction domain.

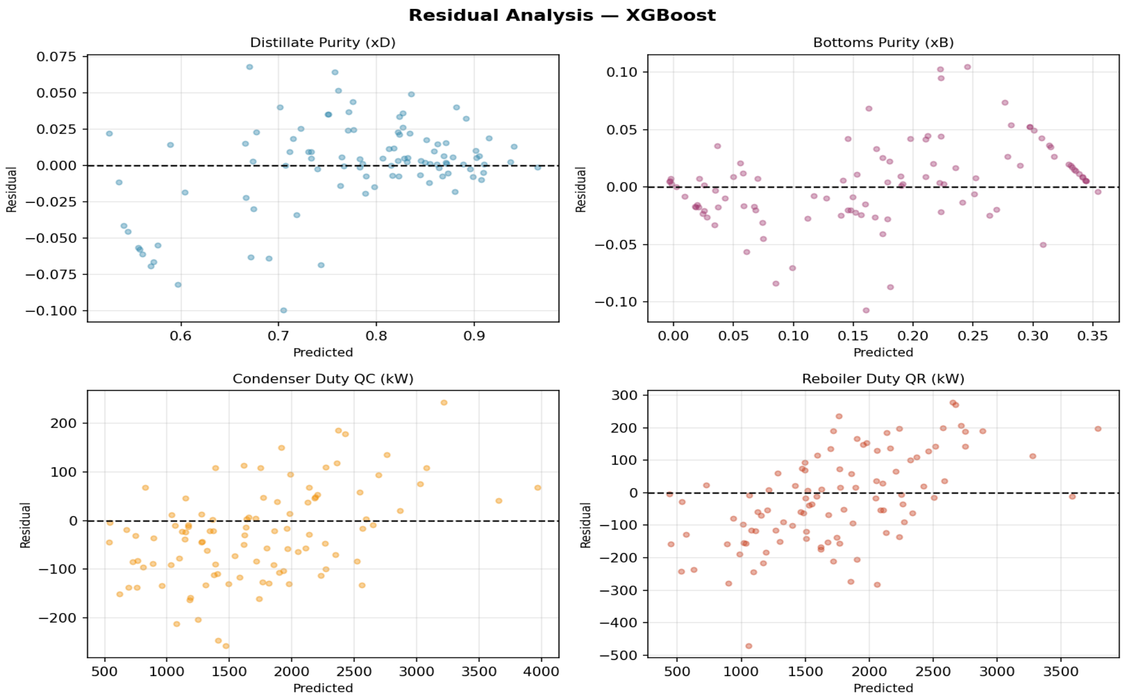

Figure 7. Residual analysis — XGBoost.

9. Overfitting & Robustness Discussion

• **XGBoost:** Train/test R^2 gap <0.03 across all targets — excellent generalization. Subsampling regularization prevented overfitting despite 300 estimators.

• **Random Forest:** Moderate overfitting on purity targets; train $R^2 \approx 0.95$ vs test ≈ 0.56. Increasing dataset size to 1000+ samples expected to close this gap significantly.

• **ANN:** Unstable due to mixed output scales (purity 0–1 vs duty 0–4000 kW). Per-target StandardScaler on outputs would correct this in future iterations.

• **Linear Regression:** No overfitting risk; $R^2 \approx 0.85$ confirms dominant linear trends in duty variables but inadequate for nonlinear purity prediction.

10. Conclusion

XGBoost is the recommended surrogate model for the Benzene–Toluene distillation column, achieving $R^2 \geq 0.92$ on all four targets with physically consistent behavior and strong generalization to unseen operating conditions.

The surrogate can replace rigorous DWSIM simulations in optimization loops (10,000× speedup), real-time process monitoring, sensitivity studies, and controller design — demonstrating practical industrial value beyond academic benchmarking.

Future work: (1) expand dataset to 2000+ samples via DWSIM batch automation, (2) apply output normalization for ANN, (3) add uncertainty quantification via conformal prediction or Gaussian Process overlay, (4) extend to ternary systems.

References

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[2] Underwood, A.J.V. (1948). Fractional distillation of multicomponent mixtures. *Chem. Eng. Prog.*, 44(8).

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[6] FOSSEE / DWSIM Team, IIT Bombay. https://dwsim.org